جامعة بيرزيت

BIRZEIT UNIVERSITY

Faculty of Engineering and Technology

Electrical and Computer Engineering Department

Digital Systems ENCS2340

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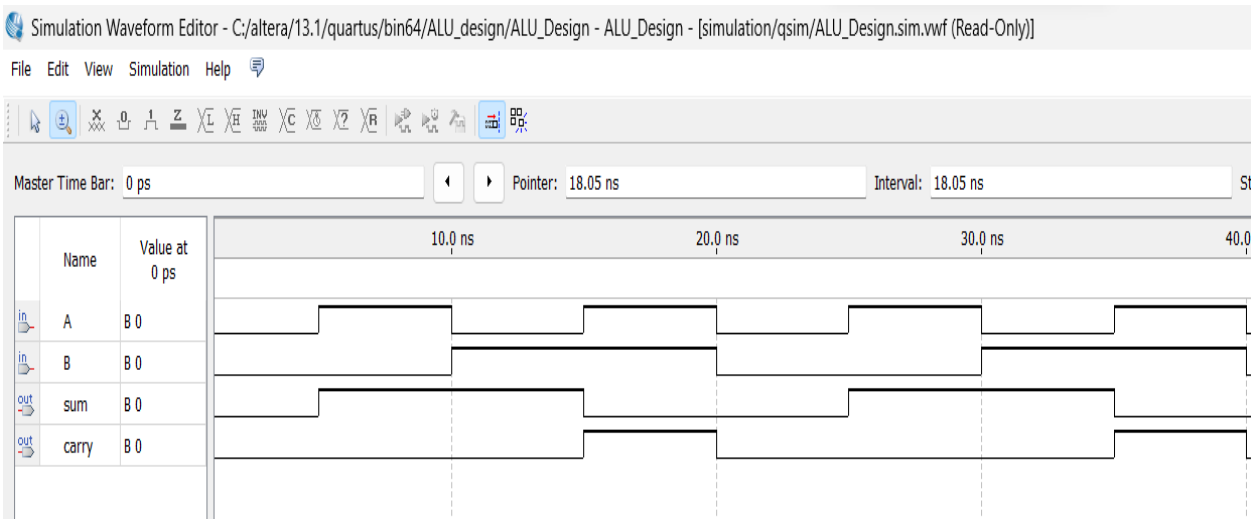
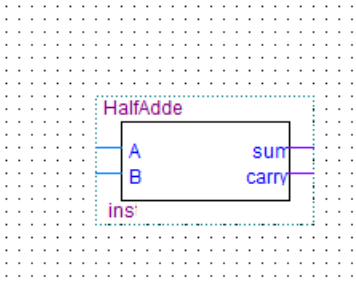
Section: 6

## HalfAdder:

```
module HalfAdder(output sum,carry ,input A,B) ;
xor g1(sum,A,B);
and g2(carry,A,B);
endmodule
```

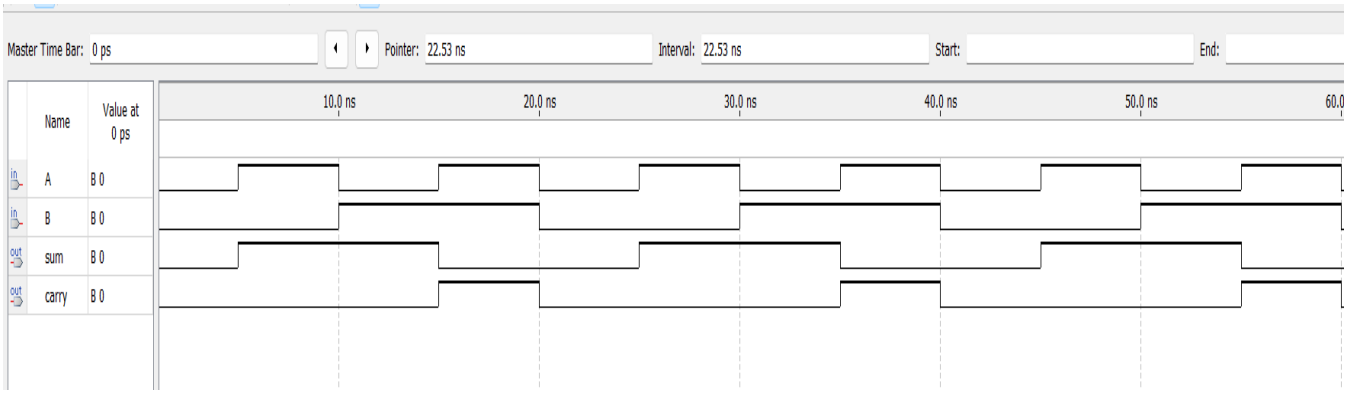


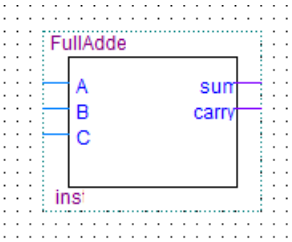

The simulation to cheack if it waork write or not



# FullAdder

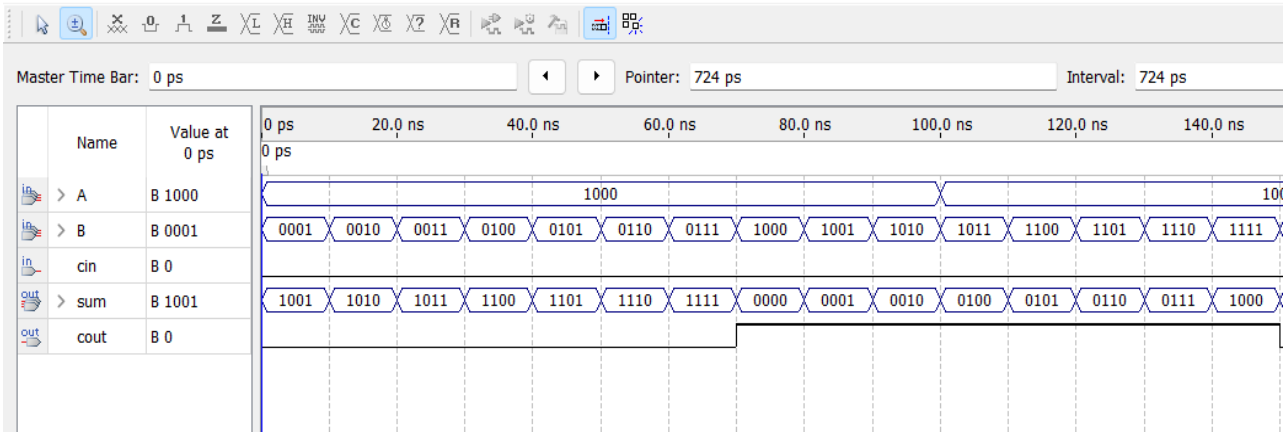
```
module FullAdder(output sum,carry,input A,B,C);
wire w1,w2,w3;
HalfAdder HA0(w1,w2,A,B);// the fulladder design form two half adder and or gate
HalfAdder HA1(sum,w3,w1,C);
or g1(carry,w3,w2);
endmodule
```

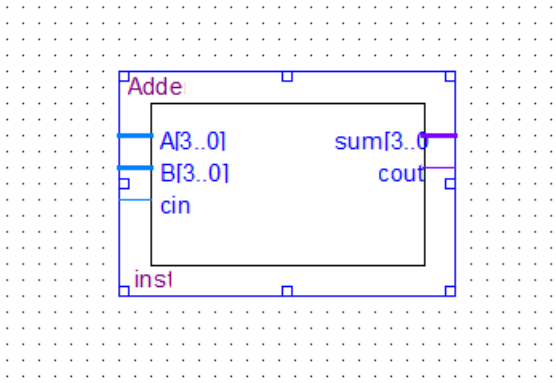


# Adder:

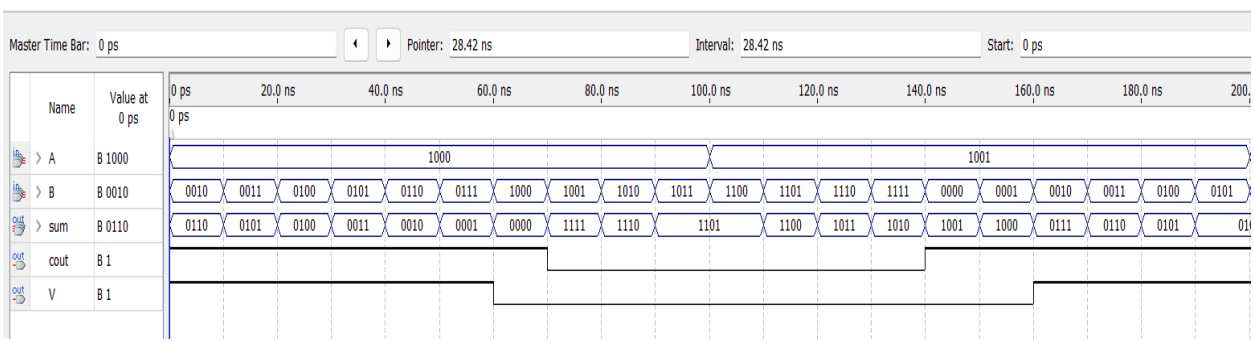
```
module Adder( output[3:0]sum,output cout,input[3:0] A,B,input cin );
wire c1,c2,c3 ;
FullAdder FA0(sum[0],c1,A[0],B[0],cin);//the adder design from 4 full adder
FullAdder FA1(sum[1],c2,A[1],B[1],c1);
FullAdder FA2(sum[2],c3,A[2],B[2],c2);
FullAdder FA3(sum[3],cout,A[3],B[3],c3);
endmodule
```

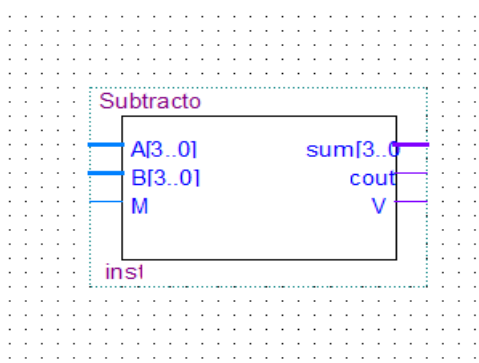


# Subtract:

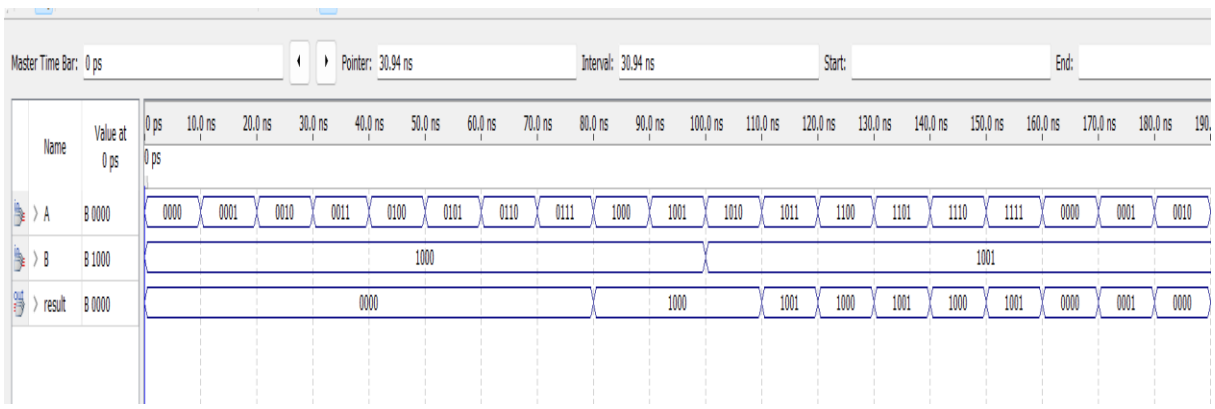
```
module Subtractor(output[3:0] sum,output cout,V,input[3:0] A,B);

wire c1,c2,c3;

wire [3:0] xB;

xor xr0(xB[0],B[0],1'b1);//the xB is the XOR btween bnd m

xor xr1(xB[1],B[1],1'b1);

xor xr2(xB[2],B[2],1'b1);

xor xr3(xB[3],B[3],1'b1);

FullAdder FAS0(sum[0],c1,A[0],xB[0],1'b1);// the subtractor design from 4 full adder with Mode =1

FullAdder FAS1(sum[1],c2,A[1],xB[1],c1);

FullAdder FAS2(sum[2],c3,A[2],xB[2],c2);

FullAdder FAS3(sum[3],cout,A[3],xB[3],c3);

xor ofl(V,cout,c3);//the over flow is the c3^cout

endmodule
```

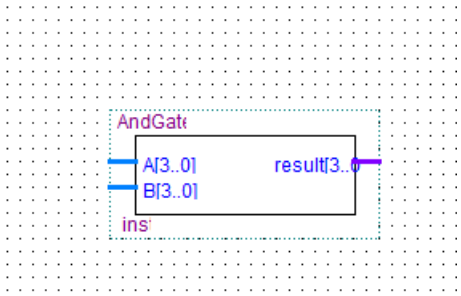


AndGate:

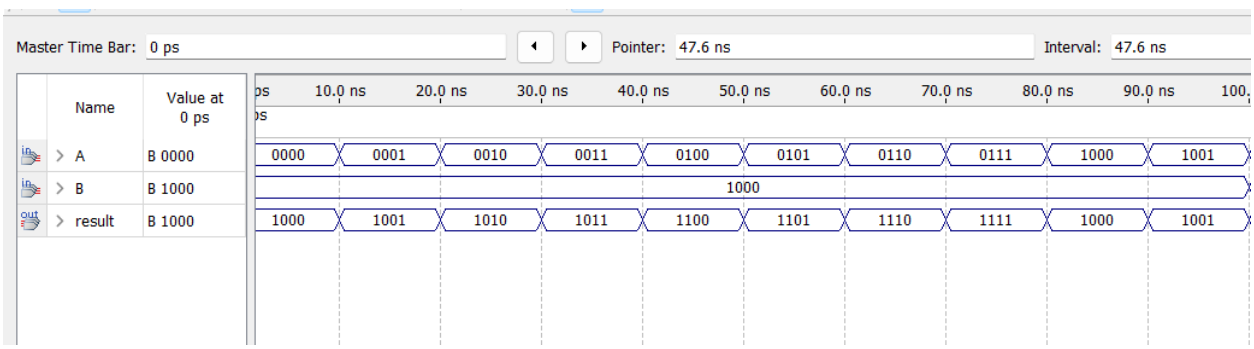
```
module AndGate(output[3:0]result, input [3:0] A,B );

and ANG0(result[0],A[0],B[0]);

and ANG1(result[1],A[1],B[1]);

and ANG2(result[2],A[2],B[2]);

and ANG3(result[3],A[3],B[3]);

endmodule
```

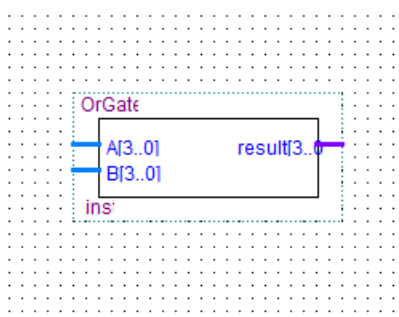


## OrGate:

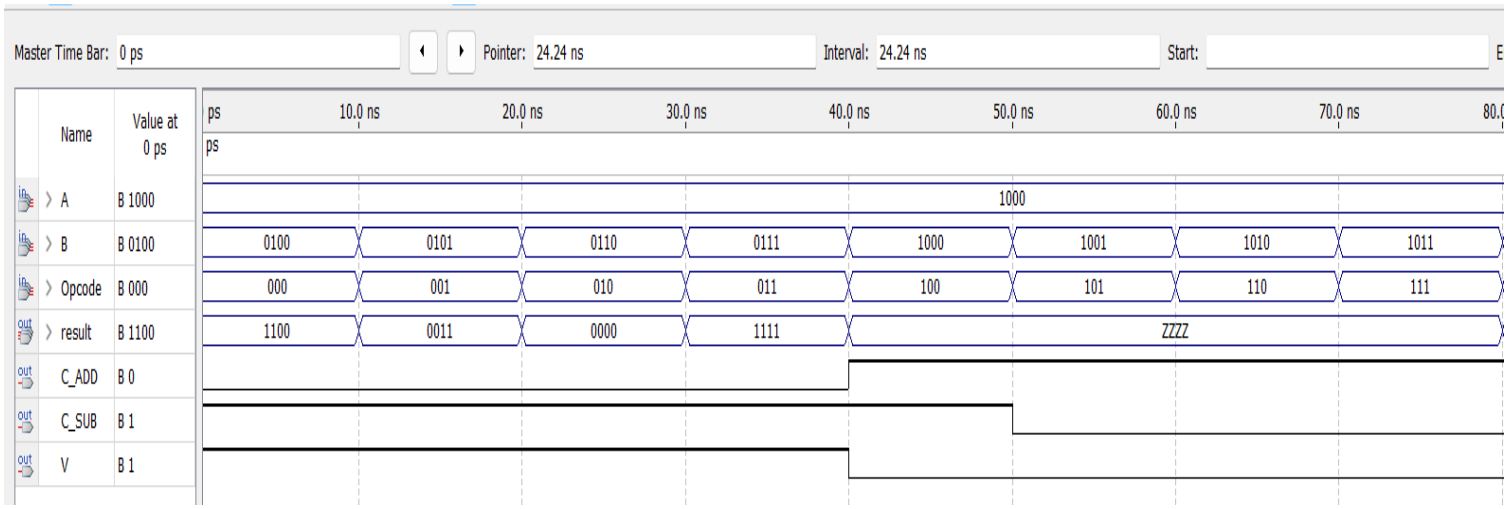
```
module OrGate(output [3:0] result,input[3:0] A,B);
or ORG0(result[0],A[0],B[0]);
or ORG1(result[1],A[1],B[1]);
or ORG2(result[2],A[2],B[2]);
or ORG3(result[3],A[3],B[3]);
endmodule
```

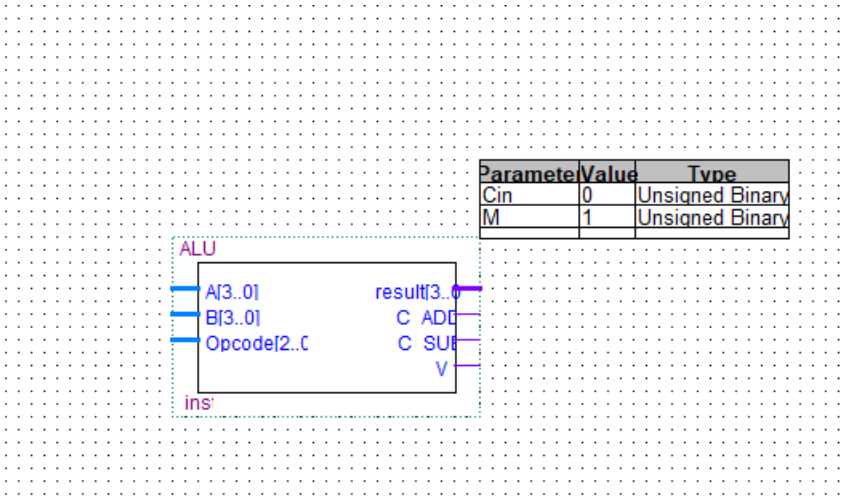


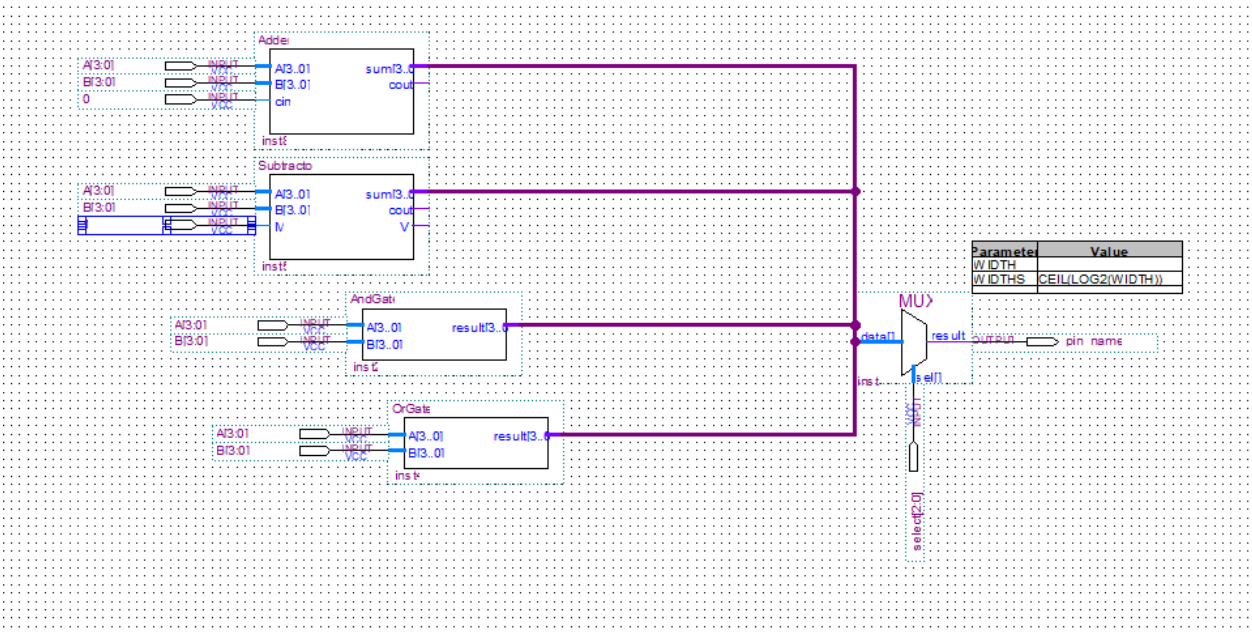
## ALU:

```
module ALU(output reg [3:0] result,output C_ADD,C_SUB, V , input [3:0] A,B, input [2:0] Opcode);

//i make select 3-bit because the assignment ask for 3-bit , it can work by 2-bit select only .

parameter Cin=1'b0, M=1'b1;

wire [3:0] I_or,I_and, I_Adder,I_Subctract;

//wire Carry ;

OrGate IORG(I_or,A,B) ;// I_or will give u the answer for the A|B

AndGate IAND(I_and,A,B);//I_AND will give u the answer for the A*B

Adder IAdd(I_Adder,C_ADD,A,B,Cin) ;// I_Adder will give u the answer for the A+B where  cin=0

Subtractor Isub(I_Subctract,C_SUB,V,A,B,M);//I_Subctract will give u the answer for the A-B where Mode =1


always @(Opcode or A or B)//alwas if there is chang on slect or A ,B enter the loop

begin//start the loop

case(Opcode)// when we have change on the select

3'b000: result =I_Adder;//IF C

3'b001: result = I_Subctract ;

3'b010: result =I_and;//if the change was 10 then go to the ANDGate

3'b011: result =I_or;//if the chang was 11 then go to the Orgate

default: result= 4'bzzzz;//highimpedance or we dont need answer when we have other value for select // if we but the select 2 bit we don't need it

endcase


end

endmodule
```


Master Time Bar: 0 ps
Pointer: 24.24 ns
Interval: 24.24 ns
Start:
Name
Value at 0 ps
ps
10.0 ns
20.0 ns
30.0 ns
40.0 ns
50.0 ns
60.0 ns
70.0 ns
A B 1000
B B 0100
Opcode B 000
result B 1100
C_ADD B 0
C_SUB B 1
V B 1
1000
0100 0101 0110 0111 1000 1001 1010 1011
000 001 010 011 100 101 110 111
1100 0011 0000 1111 ZZZZ


Parameter Value Type
Cin 0 Unsigned Binary
M 1 Unsigned Binary
ALU
A[3..0]
B[3..0]
Opcode[2..0
result[3..0
C_ADD
C_SUB
V
inst

## How the circuit looks like




Conclusions:

We conclude that we can design or construct any circuit using computers and this will enable us to save time, effort, and money and this will also enable us to design or construct more difficult circuits.